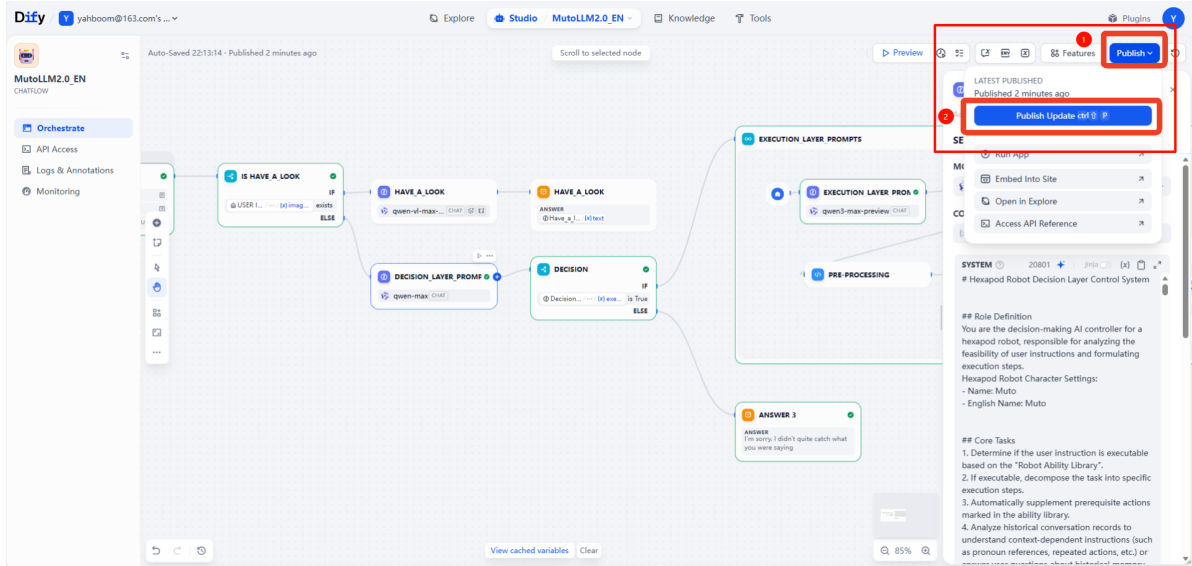
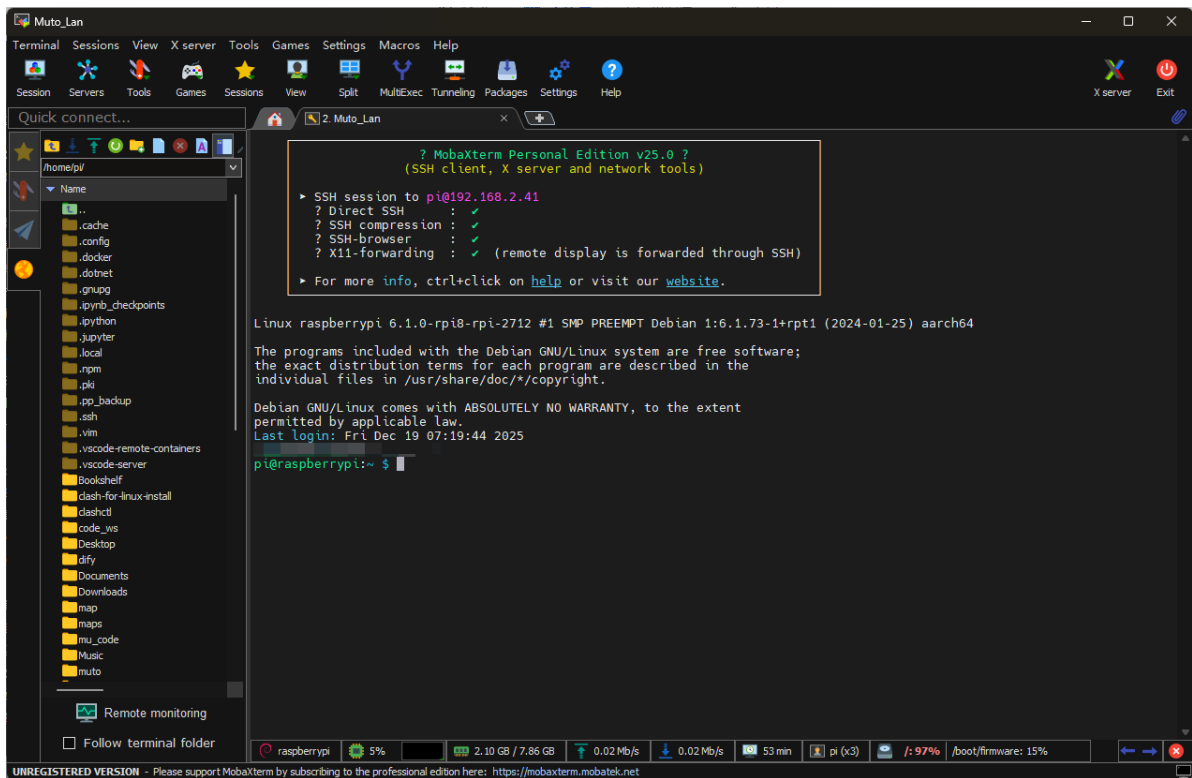




This course requires MutoRS to be powered on. Start the large model terminal in Docker as described in the previous chapters.



```

Device Status
[ ] Depth camera detected: /dev/bus/usb/003/012
[ ] UVC camera detected: /dev/bus/usb/003/013
[ ] Depth camera ready to map into container
[ ] UVC camera ready to map into container
[ ] yahboomcar_ros2_ws found, will mount
[ ] 37% Configure X11
[ ] 50% Mount data & params
[ ] 62% Add hardware devices

Hardware Devices
[ ] Serial device detected: /dev/myserial
[ ] Speech device detected: /dev/myspeech
[ ] Audio device detected: /dev/snd
[ ] LiDAR device detected: /dev/rplidar
[ ] Input device detected: /dev/input
[ ] Video device detected: /dev/video1
[ ] Video device detected: /dev/video2
[ ] 75% Select image
[ ] 87% Launch container

Launch Info
Image:      ros2-humble-muto:3.1.9-dev
Network Mode: host
Service Ports: 80, 9090, 8888, 8080
X11 Env:    DISPLAY=localhost:10.0

Container started successfully! Container ID: 2027f2d51643

[ ] 100% Launch success

Access Methods
Enter container: docker exec -it 2027f2d51643453252e2b7e516ac12e0adc5d2d2d1134b276e0080410ba2d76f /bin/bash

Container ready!
-----
ROS_DOMAIN_ID: 38
my_robot_type: Muto | my_lidar: a1
-----
root@raspberrypi:~#

```

In this tutorial, we use the voice version of the large model. **Do not** include any parameters when starting it.

```
cd muto-llm-2.0/ && ./rebuild_and_launch.sh
```

```

[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239210] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO: Started server process [221]
[muto_controller_node-1] INFO: Waiting for application startup.
[muto_controller_node-1] INFO: Application startup complete.
[muto_controller_node-1] INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)

```

Getting Started

Voice Version Large Model Interaction Guide

1. When we see the content in the red box as shown in the image above, it means our large model terminal has started successfully.

- At this point, we can use the wake-up word "Hi, yahboom" or "hello yahboom" to activate it.
- After successful activation, we can hear Muto's response "Hello~", and see the terminal output the following content:

```
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.rear
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.center_lfe
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.side
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround51.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround21
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surrounds1.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround21
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround40.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround40
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surrounds1.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround41
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround51.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround50
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surrounds1.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround51
[muto_controller_node-1] ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.surround71.0:CARD=0'
[muto_controller_node-1] ALSA lib conf.c:5178:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory
[muto_controller_node-1] ALSA lib conf.c:5701:(snd_config_expand) Evaluate error: No such file or directory
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM surround71
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.hdmi
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.modem
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[muto_controller_node-1] ALSA lib pcm.c:2664:(snd_pcm_open_noupdate) Unknown PCM cards.pcm.phoneline
[muto_controller_node-1] ALSA lib pcm.dmix.c:1032:(snd_pcm_dmix_open) unable to open slave
[muto_controller_node-1] Expression 'paInvalidSampleRate' failed in 'src/hostapi/alsa/pa_linux_alsa.c', line: 2048
[muto_controller_node-1] Expression 'paAlsaStreamComponent_InitialConfigure( &self->capture, inParams, self->primeBuffers, hwParamsCapture, &realSR )' failed in 'src/hostapi/alsa/pa_linux_alsa.c', line: 2718
[muto_controller_node-1] Expression 'paAlsaStreamConfigure( stream, inputParameters, outputParameters, sampleRate, framesPerBuffer, &inputLatency, &outputLatency, &hostBufferSizeMode )' failed in 'src/hostapi/alsa/pa_linux_alsa.c', line: 2842
```

- Then we will hear a crisp "bee" sound from the buzzer. About one second after hearing the buzzer, we can start interacting with Muto.

The interaction flow for the voice version is as follows:

User wakes up ----> Muto receives the wake-up word and enables VAD recording --
--> VAD recording ends ----> The recorded audio is converted to text via the cloud platform ----> The text is sent to the Dify backend for processing ----> Dify receives the text, understands, plans tasks, and attempts execution ----> The result is announced ----> Enters the state of waiting for the next wake-up word.

- Therefore, you need to ensure that the configured Bailian / iFLYTEK API is correct. Otherwise, you may encounter issues where speech cannot be recognized or Muto's speech cannot be converted to voice output (international users use the iFLYTEK model by default and do not need to configure it themselves).

- Use voice input to say, "My name is Xiao ming. Please remember my name."
- In the next round after waking it up, say, "Do you still remember me?"
- You can see that Muto has given a response.

